

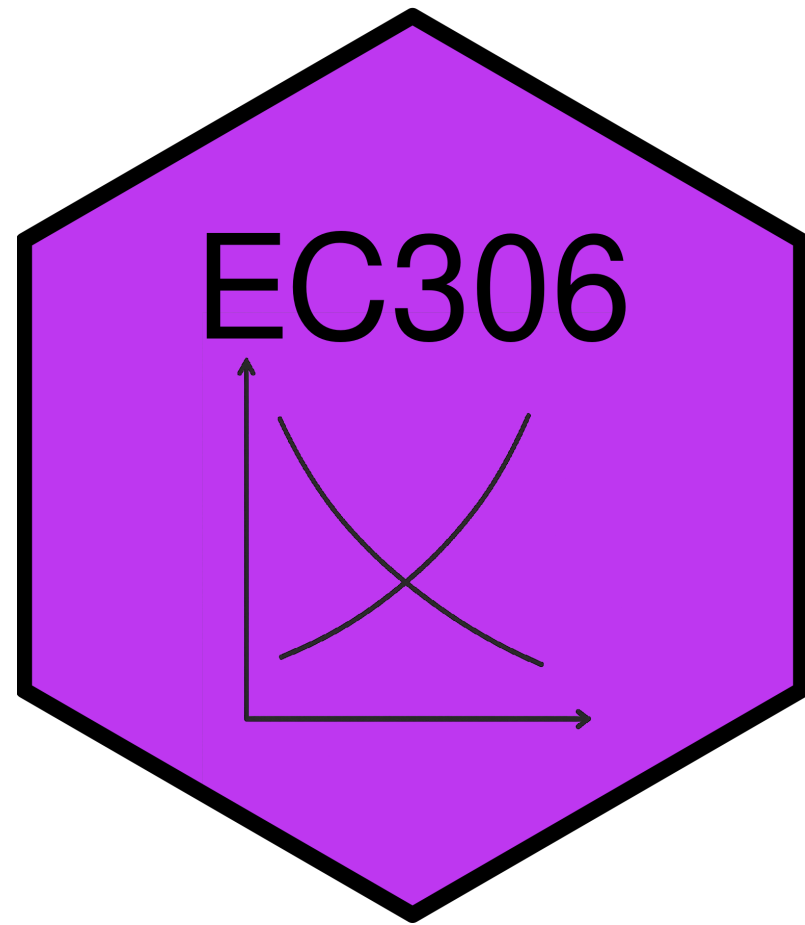
Discrimination and Male-Female Earnings Differentials

EC306- Wages and Employment

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Goals of This Section

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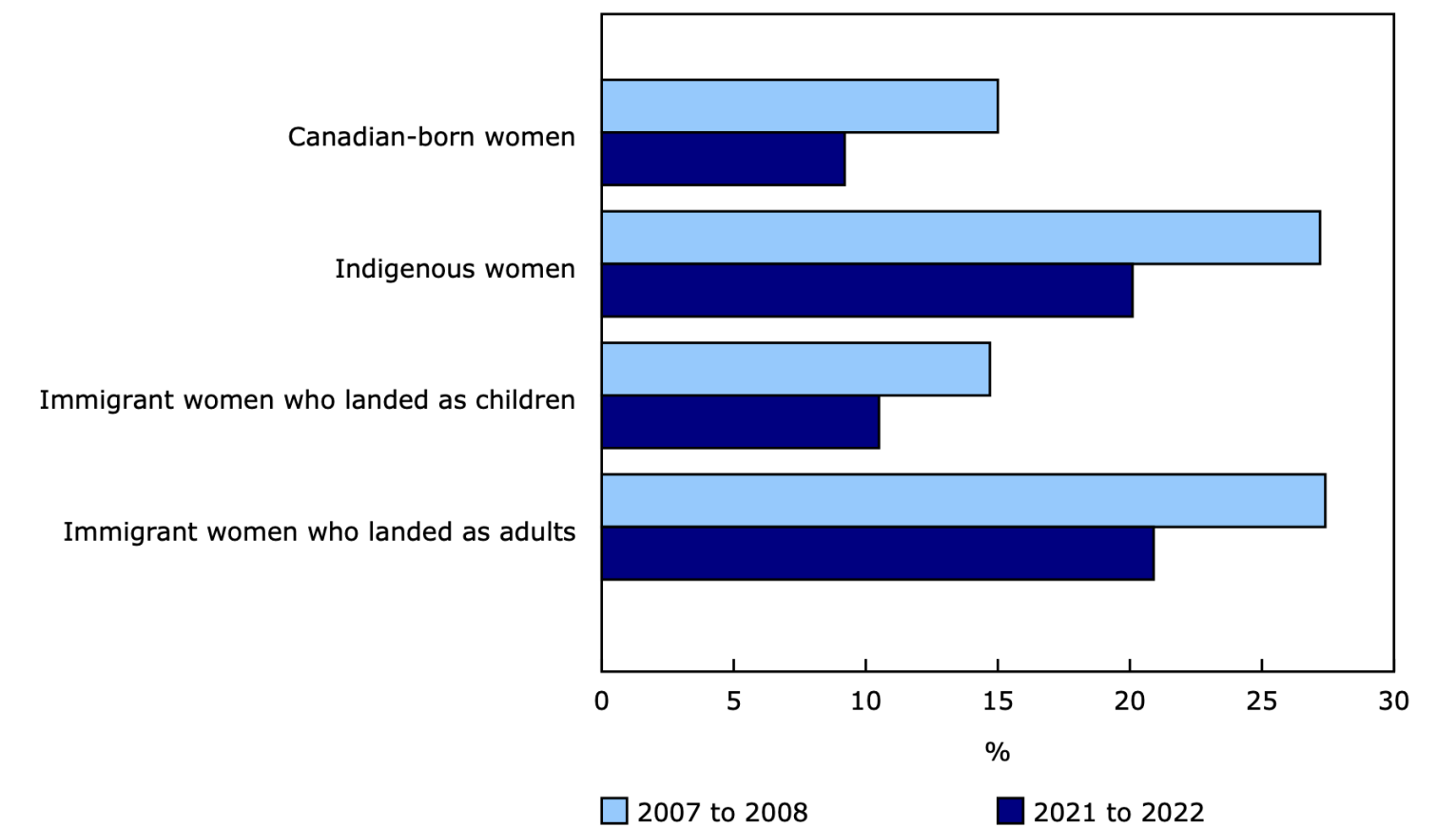
- Discuss trends in educational attainment in Canada
- Introduce Human Capital Theory of education investment
- Introduce Signalling Model of education investment
- Discuss empirical methods to estimate returns to schooling
- Review empirical evidence on returns to schooling
- Discuss job training

Introduction

Male and Female Earnings

Male and Female Earnings

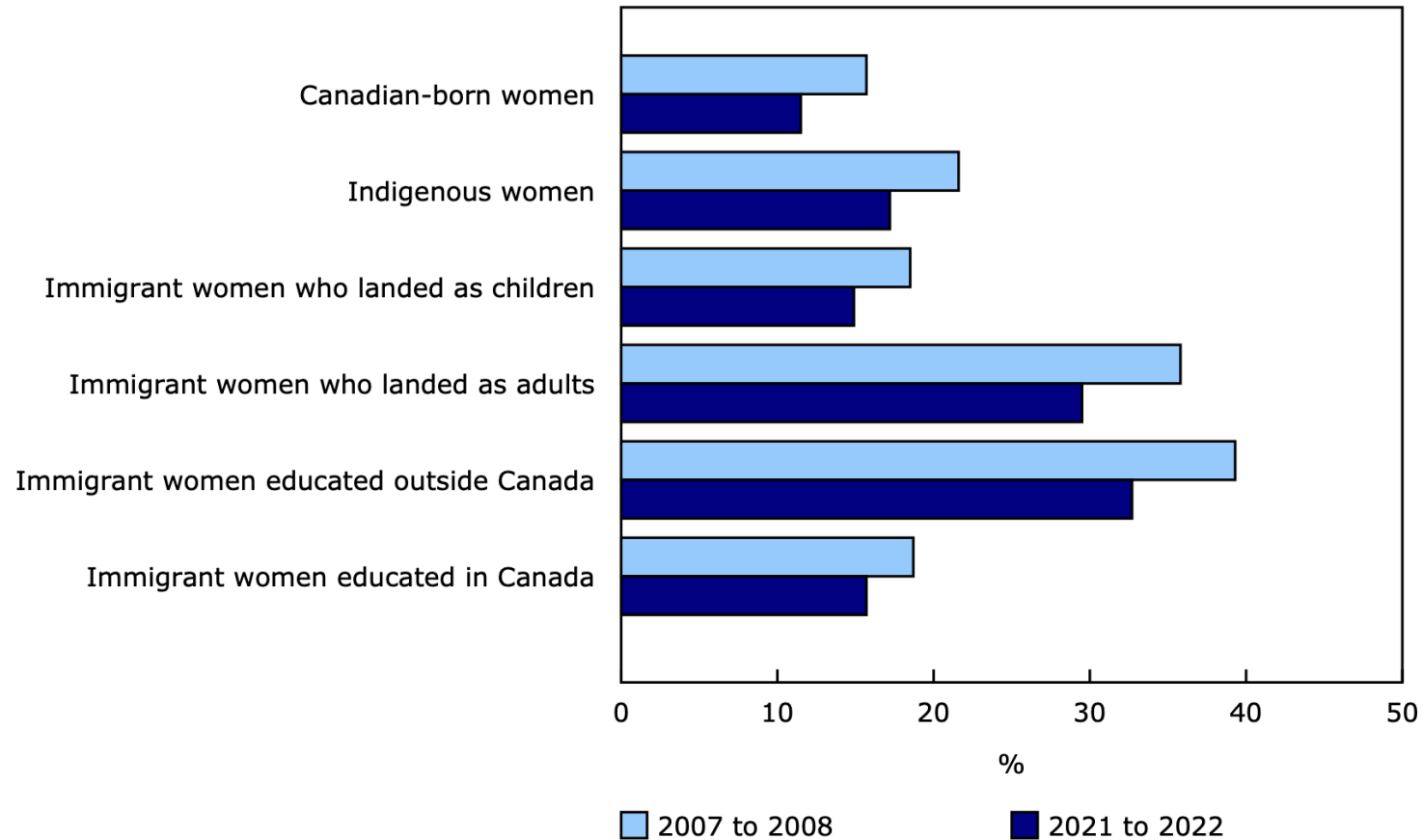
Gap in average hourly wage relative to Canadian-born men, by group of women, 2007 to 2008 and 2021 to 2022



Note(s): "Canadian-born" refers to non-Indigenous workers born in Canada.
Source(s): Labour Force Survey (3701), March and September monthly files, 2007, 2008, 2021, and 2022.

Male and Female Earnings

Gap in average hourly wage relative to Canadian-born men among workers holding a bachelor's degree or higher, 2007 to 2008 and 2021 to 2022



Note(s): "Canadian-born" refers to non-Indigenous workers born in Canada.

Source(s): Labour Force Survey ([3701](#)), March and September monthly files, 2007, 2008, 2021, and 2022.

Male and Female Earnings

- Clearly men and women earn different amounts on average
- There are two broad explanations for this gap:
 1. Differences in characteristics (education, experience, occupation, industry, hours worked, etc.)
 2. Differences in the returns to those characteristics (discrimination)
- Our models so far have not left room for discrimination
 - We treated all workers as the same
 - So there should not be any differences in returns

Male and Female Earnings

- In this section, we explore economic theories of discrimination
 - Can come from employers, customers, co-workers
 - Can be statistical or taste-based
- It is difficult empirically to estimate discrimination
 - Need to control for all relevant characteristics
 - Need to separate out differences in characteristics from differences in returns
- There are methods that can help us estimate discrimination
 - Oaxaca-Blinder decomposition
 - Audit studies
 - Experiments

Demand Theories of Discrimination

Demand-Side Theories

- Demand theories focus on the behavior of employers in demanding labor
- Can happen for various reasons:
 - Pure prejudice
 - Customer preference to buy from male workers
 - Male aversion to working with women
- These all cause a fall in the demand for women relative to men in the labour market
 - Reduces wages and employment of female workers

Taste-Based Discrimination

- Theory of taste-based discrimination pioneered by Becker (1957)
- Imagine that firms produce according to the production function:

$$y = f(N_m + N_w)$$

- With N_m indicating the number of male workers, and N_f indicating the number of female workers
- So in a competitive market their profits are

$$\pi = p \cdot f(N_m + N_w) - w_m N_m - w_w N_w$$

- With w_m indicating the wage of male workers, and w_f indicating the wage of female workers

Taste-Based Discrimination

- Firms do not maximize profit strictly, but instead maximize a utility function

$$U = \pi - d \cdot N_w$$

- Utility is partly a function of profit
- But also includes d , which measures the degree of discrimination
 - if $d = 0$, no discrimination and maximizing utility is the same as maximizing profit
 - if $d > 0$ firms get a disutility from hiring women

Taste-Based Discrimination

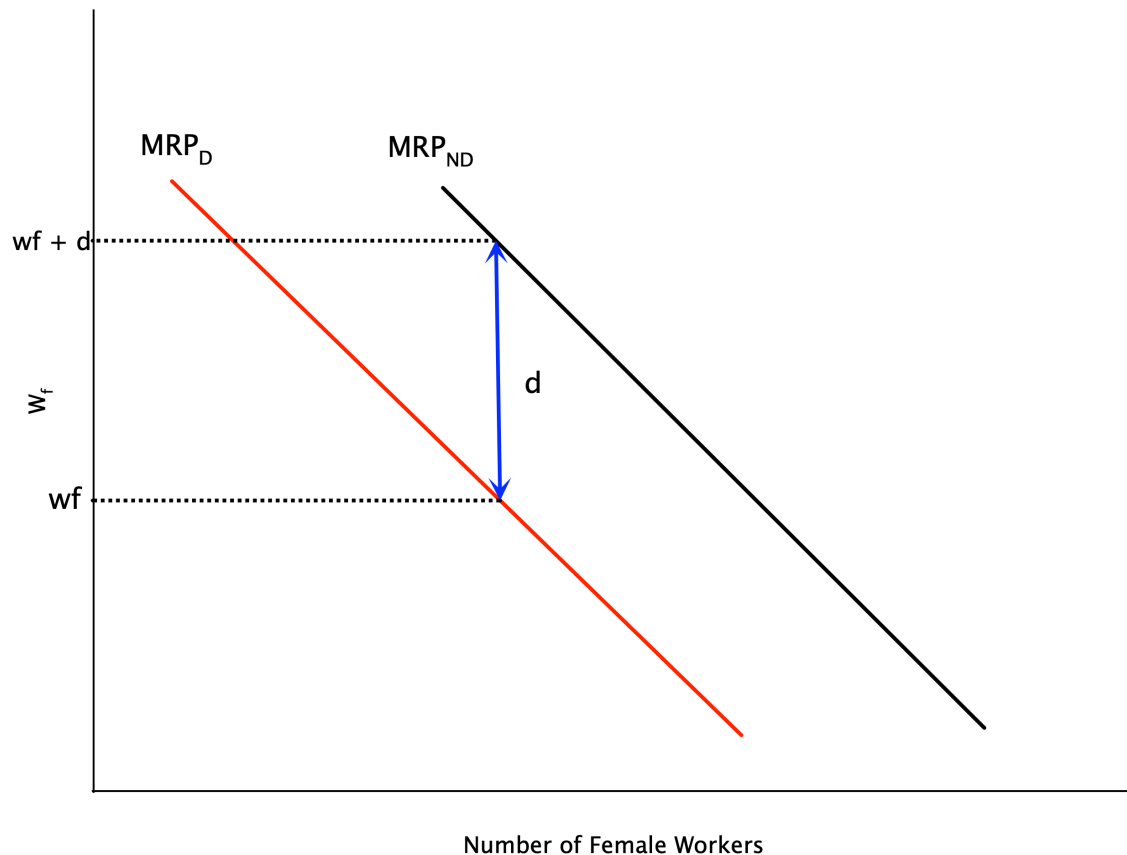
- The first order conditions for utility maximization are:

$$p \cdot MP_{N_m} = w_m$$

$$p \cdot MP_{N_w} = w_w + d$$

- For both men and women, firms hire up to the point where the value of the marginal product equals the cost of hiring
 - But for women, there is an additional cost d due to discrimination
- If $w_m = w_f$, and men and women are equally productive, this implies the firm hires less women
 - Discrimination drives a wedge between wages and the MRP

Taste-Based Discrimination

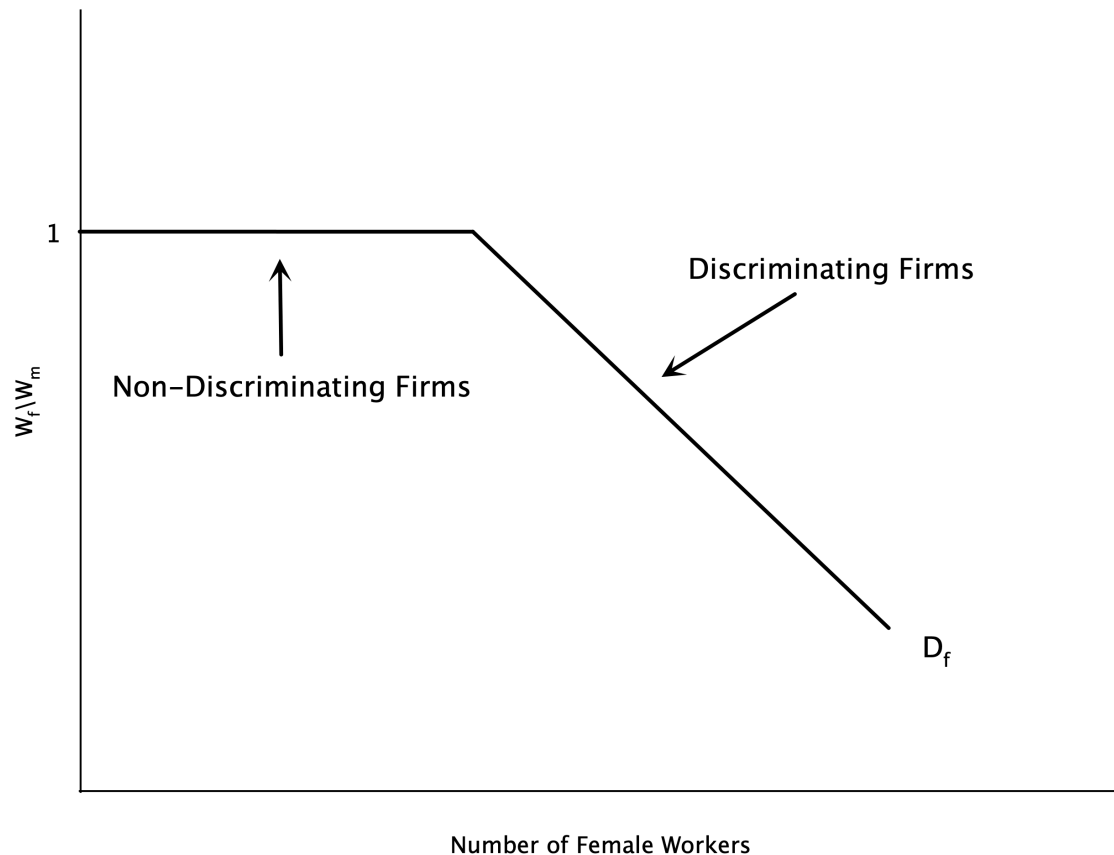


- Graph plots labour demand for women
- Discrimination shifts demand curve left
- Without discrimination, firm hires where $w_f = MRP_{ND}$
 - ND = non-discriminatory
- With discrimination, firm hires where $w_f + d = MRP_D$
 - Or equivalently $w_f = MRP_D - d$
- Implies a lower level of female labour

Taste-Based Discrimination

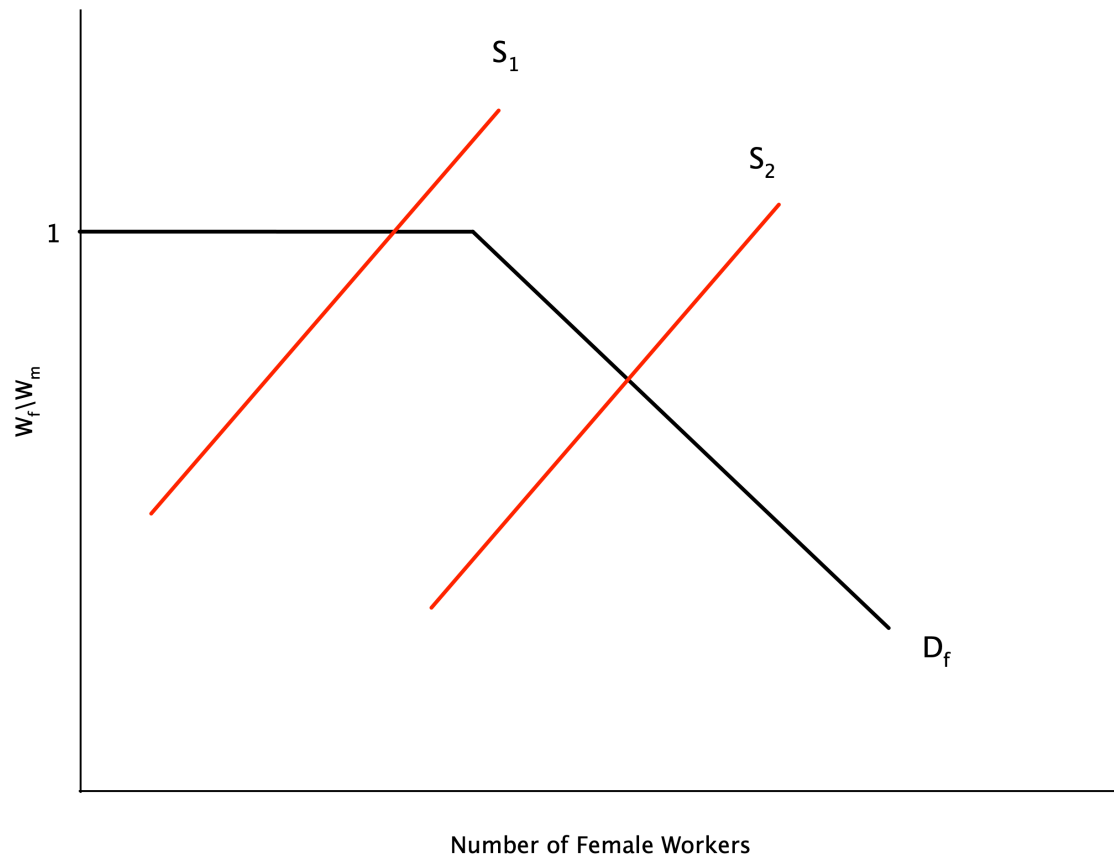
- Above analysis is for a single employer
- Employers discriminate along a spectrum
 - Some have a high d , some have a low or no d
- Suppose women are able to seek out the non-discriminating employers
- If we order the firms from lowest to highest by the amount of discrimination
 - We can get the market demand curve by summing the individual demand curves horizontally
- This demand curve is plotted on the next slide

Taste-Based Discrimination



- Graph plots demand for female labour in the market
- On the vertical axis is the ration of female to male wages
- For non-discriminating firms, this ratio is 1
 - A certain set of firms will hire women at this ratio
 - This is the flat portion of the demand curve
- As the supply of women grows, they must work at more discriminating firms
 - These firms will only hire when the wage ratio falls

Taste-Based Discrimination



- If the supply of female labour is small enough, they can work at non-discriminating firms
 - Wage ratio is 1
- With a larger supply, some will have to work at discriminating firms
 - Wage ratio falls below 1
 - This is where the demand curve slopes downwards
 - Wage gap emerges

Statistical Discrimination

- Taste-based discrimination relies on employers having a “taste” for discrimination
 - Overt prejudice
- Some discrimination may happen because of imperfect information
 - Employers do not know the productivity of workers perfectly
- Instead they use signals to infer worker quality
- With **statistical discrimination** firms use group characteristics to infer individual productivity
 - If one group has lower average productivity, firms may pay them less on average

Statistical Discrimination

Theory

- Assume that the abilities of male and female workers are distributed with means $\bar{a}_m$ and $\bar{a}_f$
- Suppose the firm has access to a signal of ability x for each person
- Actual ability is not known with certainty, but expected ability is

$$a(x) = \bar{a} + \alpha(x - \bar{a})$$

- where α is the reliability of the signal
 - If $\alpha = 0$, the signal provides no information beyond the average
 - If $\alpha = 1$, the signal is perfectly informative
- Workers are paid a wage equal to their marginal revenue product

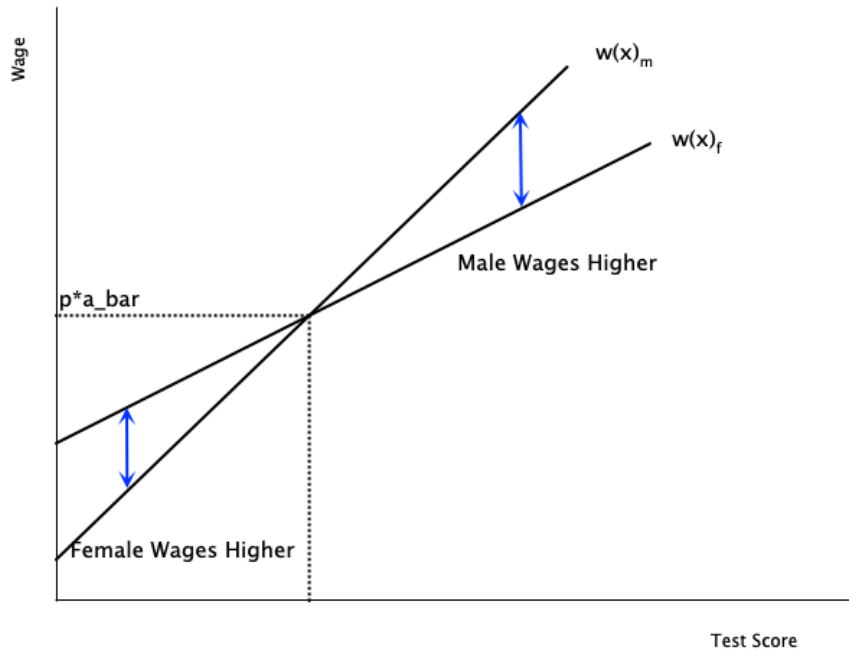
Statistical Discrimination

- Paying workers their marginal revenue product implies

$$w(x) = pa(x) = p\bar{a} + p\alpha(x - \bar{a})$$

- To analyze discrimination, assume $\bar{a}_m = \bar{a}_f$
 - Men and women are equally productive on average
- Assume the signal is more reliable for men than for women
 - If $\alpha_m > \alpha_f$
- Differences arise between workers who score the same on the signal
 - If a man and woman score above average ($x - \bar{a} > 0$), then $w(x)_m > w(x)_f$
 - If a man and woman score below average ($x - \bar{a} < 0$), then $w(x)_m < w(x)_f$

Statistical Discrimination



- In this graph we have wages as a function of the signal
- The male line is steeper because the signal is more reliable ($\alpha_m > \alpha_f$)
- For test scores above average, men are paid more
- For test scores below average, women are paid more
- At the average, they are paid the same

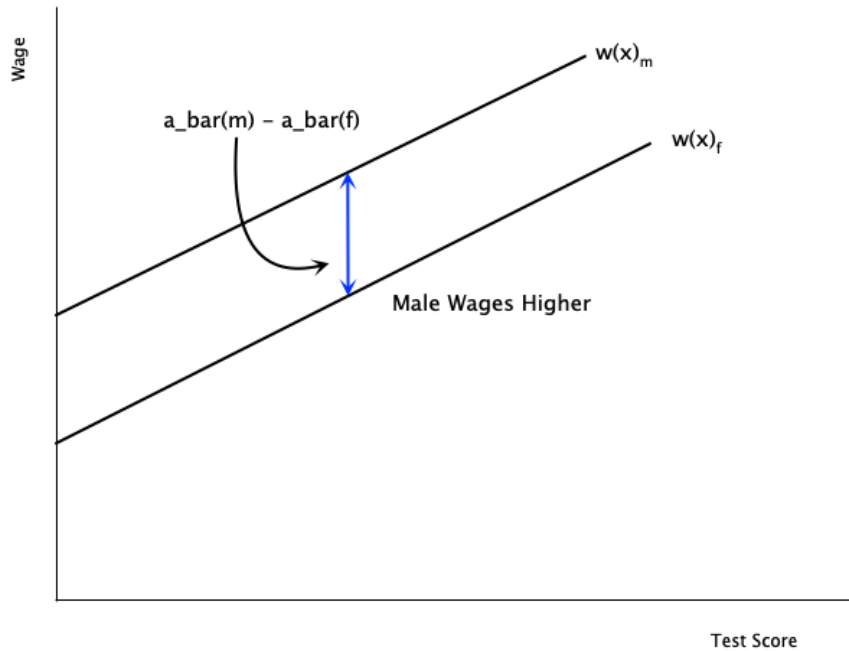
Statistical Discrimination

- There are no underlying differences between males and females
 - Differences in wages arise because of differences in the quality of the signal
 - Firms place more weight on average ability for female workers
 - Firms place more weight on the actual signal for male workers
- At low levels of the signal, women are paid more than men
- But where does the assumption $\alpha_m > \alpha_f$ come from
 - Possibly related to prejudicial attitudes of firms
 - Some believe standardized tests favour males

Statistical Discrimination

- There may also be differences in $\bar{a}_m$ and $\bar{a}_f$
 - Possibly due to pre-market discrimination leading to lower human capital among women
- In this case, males are paid more at every signal level
- Wage differences do not necessarily arise from overt discrimination
 - They may originate in differences in information signals
 - But it is difficult to explain why such differences in signal quality exist

Statistical Discrimination



- In this graph the quality of the signal is the same
- Slopes of the lines are the same
- But there are differences in average productivity
 - Male line lies above female line
- Men will be paid more because of the average difference

Supply Theories of Discrimination

Supply-Side Theories

- We discussed demand-side theories of discrimination
 - Focused on employer behaviour
- Discrimination may also arise on the labour supply side
 - Factors that affect where women choose or are able to work

Crowding Hypothesis

- Older theory emphasizing that women are overrepresented in certain occupations
- This pushes up their labour supply in those occupations
 - Which depresses wages relative to men
- Discrimination comes from the supply side
 - Women are paid their marginal product
 - But are corralled into a small set of jobs, which depresses wages

Dual Labour Market Theory

- Similar in spirit to the crowding hypothesis
- Theory says there are two labour markets
 - Primary market
 - High wages, good working conditions, opportunities for advancement
 - Unionized, stable, expanding
 - Secondary market
 - Low wages, poor working conditions, little opportunity for advancement
 - Non-unionized, highly competitive, declining
- Theory posits that women are concentrated in the secondary market
- Prejudice prevents women from entering primary market

Lower Reservation Wages

- One theory is that female reservation wages are lower because
 - Costly to search for non-discriminating employers
 - Job search is limited by strong household ties
 - Ability to negotiate higher pay limited by lower threat of leaving
- With a lower reservation wage, women are more likely to be paid less
 - Contributes to pay gap
- Some evidence of this in empirical studies

Risk Aversion

- Large literature documenting that women are less aggressive/competitive in bargaining
- Also less confident in their abilities
- Evidence of this in many studies in economics and psychology
- Might mean women are less likely to compete for higher paying jobs

Differences in Preferences

- May be differences in preferences for
 - Education
 - Training
 - Hours
 - Work conditions
 - Occupation
- All of these are correlated with pay
- Issue is that these choices are sometimes a product of discrimination
 - STEM fields are an example
 - History of discrimination in this field may lead women not to choose it
- Hard to separate out preferences from discrimination effects

We will cover this in more detail in a few slides



Non-Competitive Theories of Discrimination

Non-Competitive Theories

- One issue with the above theories is that they assume competitive markets
- Men and women are assumed to have equal productivity
- Discriminating firms hire less women because of prejudice
- But non-discriminating firms can hire more women at lower wages
 - Would increase their profits
- Non-discriminating firms would bid up the wages of women
 - Should close the gap

Non-Competitive Theories

- But the gap continues to exist
- Non-competitive theories of discrimination try to explain this
- The unifying theme is that markets are not perfectly competitive
 - May be monopsony power, imperfect information, search frictions, etc.
- This can lead to persistent wage gaps

Imperfect Information

- Even if firms know that they can increase profits by hiring more women, they might not be able to immediately change workforce
- Hiring and recruiting are costly
 - There are search frictions
 - Quasi-fixed costs
- Firms may be slow to adjust to profit opportunities
- This can lead to persistent wage gaps
- But does not explain why the gaps have persisted for so long

Queueing Theory

- Some firms pay **efficiency wages**
 - Wages above the market clearing level to increase productivity
- This results in a queue of workers wanting to work at the firm
 - Due to the higher wages
- While rationing jobs, firms may discriminate

Political Pressure

- It is sometimes the case that markets decisions are overridden by political pressures
- A male-centric government or union may
 - Hire more men in government jobs
 - Be discriminatory with education and training
- The political pressure could come from interest groups
 - May have a lot of money to lobby the government

Monopsony Power

- Monopsony gives firms power to set wages
- Monopsonies may take advantage of relative immobility of women to lower their wages
 - Recall they may have stronger ties to household
- They may also try to pit men and women against each other
 - Create competition between the groups to lower wages

Systemic Discrimination

- Idea is that discrimination is embedded in social and economic systems
- Has been going on for so long, that it is now self-perpetuating
- Could come from
 - Existence of “old boys” mentality
 - Old job requirements that implicitly discriminate but continue to exist

Productivity Differences

Productivity Differences

- It may be the case that men and women have productivity differences
 - This could determine pay differences even in competitive markets
- Note: not *innate* productivity differences
- Differences coming from acquisition of human capital over time
 - Education
 - Training
 - Work experience
- If human capital investments are a choice, then pay differences are not discrimination
- But it could be that choices are influenced by discrimination

Dual Role

- Traditionally women have more of a dual role than men
 - Balance household and market work
- If this continues to be true, then women
 - Have less continuity in work (mainly from interruptions for children)
 - Are more likely to have periods of paid work and labour force withdrawals
- This leads to lower accumulation of human capital
 - Because there is less time to recoup the costs
- In some sense this is a rational choice
 - But is rooted in a society that does not accommodate enough for working mothers

Pre-Market Discrimination

- Discrimination may influence choices in
 - Education
 - Training
 - Career path
- Thus what looks like a rational choice is itself affected by discrimination
 - In the literature this is called **pre-market discrimination**
- Example: STEM jobs
 - Pay well, but are male dominated
 - Women are discouraged from entering these fields
 - Thus do not have access to the higher pay that comes with these jobs

Measuring Discrimination

Introduction

- Now that we have discussed theories of discrimination, how do we measure it?
- Difficult because in most cases discrimination is not directly observable
 - There is not a “discrimination” variable in datasets
- Can design experiments to measure discrimination
 - Audit studies
 - Field experiments
- But in most cases we have to rely on observational data

Introduction

- Measuring discrimination involves comparing people of equal productivity
 - So that any left over wage differences could be due to discrimination
- But we do not observe productivity directly
- Instead, hold constant variables related to productivity
 - Education
 - Experience
 - Occupation
 - Industry
 - Hours worked
- Then decompose wage difference into part due to characteristics and part due to other factors

Blinder-Oaxaca Decomposition

- Method to decompose differences in average earnings between any two groups into
 - Part due to differences in characteristics (explained)
 - Part due to differences in returns to those characteristics (unexplained)
- Suppose we estimate a regression of log earnings on observed characteristics for men and women
- The predicted value (which is also equal to the mean of log earnings) is

$$\ln(Y_m) = \hat{\beta}_m \bar{X}_m$$

$$\ln(Y_f) = \hat{\beta}_f \bar{X}_f$$

Blinder-Oaxaca Decomposition

- The difference in average log earnings is

$$\ln(Y_m) - \ln(Y_f) = \hat{\beta}_m \bar{X}_m - \hat{\beta}_f \bar{X}_f$$

- We can add and subtract $\hat{\beta}_m \bar{X}_f$ to get

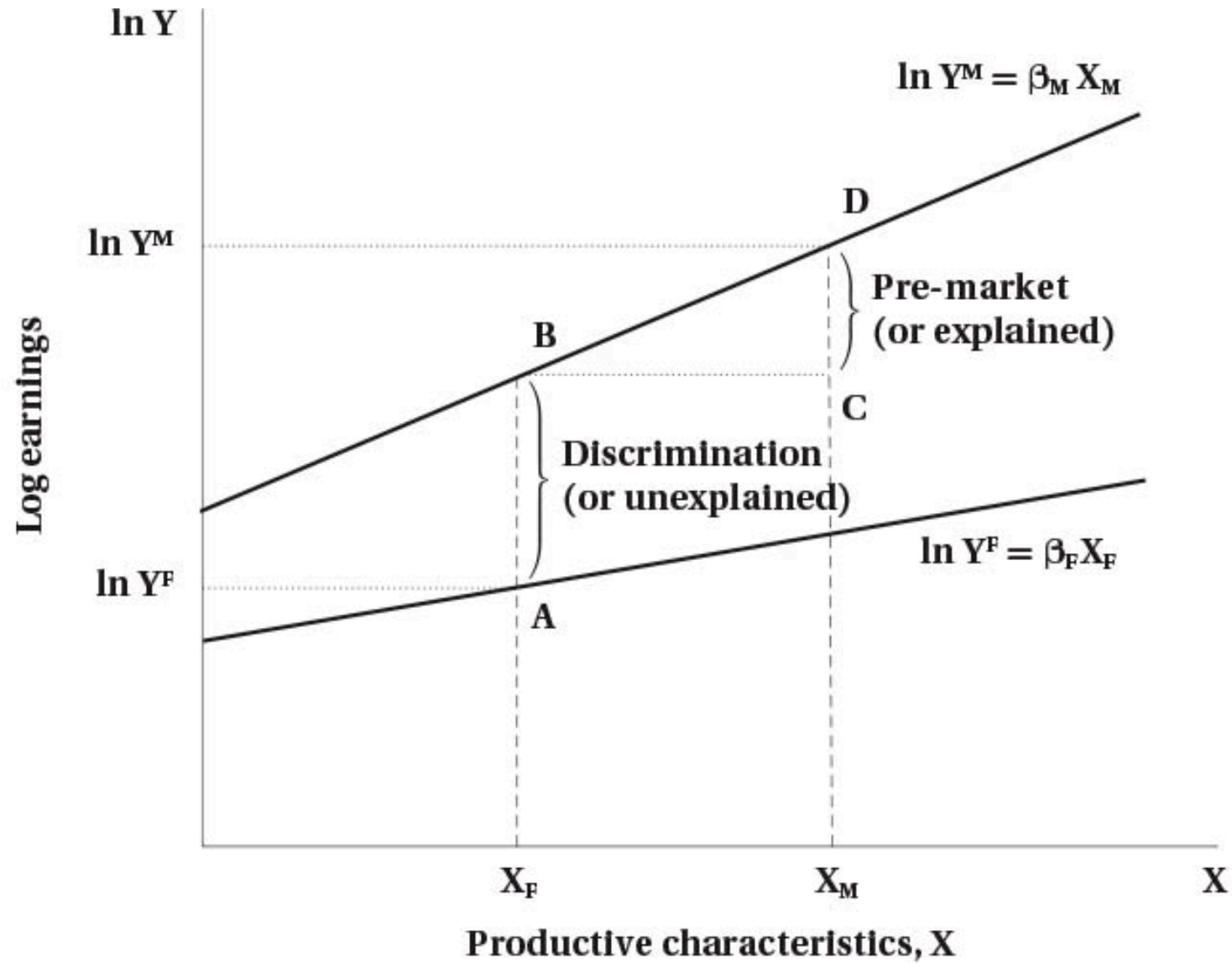
$$\ln(Y_m) - \ln(Y_f) = \hat{\beta}_m (\bar{X}_m - \bar{X}_f) + \bar{X}_f (\hat{\beta}_m - \hat{\beta}_f)$$

- Doing this allows us to separate the difference into two parts
- The first part is the explained part
 - Due to differences in characteristics
 - Holds the slope constant at $\hat{\beta}_m$ and looks at differences in average X

Blinder-Oaxaca Decomposition

- The second part is the unexplained part
 - Due to differences in returns to characteristics
 - Holds the average characteristics constant at $\bar{X}_f$ and looks at differences in slopes
 - The slopes are the returns to characteristics
- Traditionally, the unexplained part is interpreted as discrimination
 - But could also be due to unobserved productivity differences

Blinder-Oaxaca Decomposition



Blinder-Oaxaca Decomposition

- Here is an example of how this works
- Suppose we have run the following regression separately for men and women

$$\ln Y = \beta_0 + \beta_1 \text{Educ} + \beta_2 \text{Experience}$$

- The estimated coefficients are:
- Men: $\beta^M = (\beta_0^M, \beta_1^M, \beta_2^M) = (2.0, 0.10, 0.02)$
- Women: $\beta^F = (\beta_0^F, \beta_1^F, \beta_2^F) = (1.8, 0.08, 0.015)$

Blinder-Oaxaca Decomposition

- The mean of the observed characteristics are
- Men: $E\bar{\text{educ}}_M = 14$, $E\bar{\text{exp}}_M = 15$
- Women: $E\bar{\text{educ}}_F = 13$, $E\bar{\text{exp}}_F = 12$
- The average predicted log wage for men

$$\ln \bar{Y}_M = 2.0 + 0.10 \cdot 14 + 0.02 \cdot 15 = 2.0 + 1.4 + 0.3 = 3.7$$

Blinder-Oaxaca Decomposition

- The average predicted log wage for women

$$\ln \bar{Y}_F = 1.8 + 0.08 \cdot 13 + 0.015 \cdot 12 = 1.8 + 1.04 + 0.18 = 3.02$$

- The wage gap is therefore

$$\ln \bar{Y}_M - \ln \bar{Y}_F = 3.70 - 3.02 = 0.68$$

- This says that the wage gap is about 0.68 log points
 - Roughly a 68% higher wage for men

Blinder-Oaxaca Decomposition

We decompose:

$$\ln \bar{Y}_M - \ln \bar{Y}_F = \text{Explained} + \text{Unexplained}$$

- Explained (due to characteristics) component is

$$\text{Explained} = (\bar{X}_M - \bar{X}_F)\beta^M$$

- Compute differences in means:
- Education: $14 - 13 = 1$
- Experience: $15 - 12 = 3$

Blinder-Oaxaca Decomposition

- Then:

$$\text{Explained} = 1 \cdot 0.10 + 3 \cdot 0.02 = 0.10 + 0.06 = 0.16$$

- About 0.16 log points of the gap are due to men having more education and experience.
- Unexplained (slopes) component is

$$\text{Unexplained (slopes)} = \bar{X}_F (\beta_M - \beta_F)$$

- Coefficient differences:
- Education: $0.10 - 0.08 = 0.02$
- Experience: $0.02 - 0.015 = 0.005$

Blinder-Oaxaca Decomposition

- Then:

$$\text{Unexplained (slopes)} = 13 \cdot 0.02 + 12 \cdot 0.005 = 0.26 + 0.06 = 0.32$$

- Intercept difference:

$$\beta_{0M} - \beta_{0F} = 2.0 - 1.8 = 0.20$$

- This is also part of the unexplained component (baseline difference not due to X).
- Total unexplained:

$$\text{Unexplained (total)} = 0.32 + 0.20 = 0.52$$

Blinder-Oaxaca Decomposition

- The decomposition is therefore:

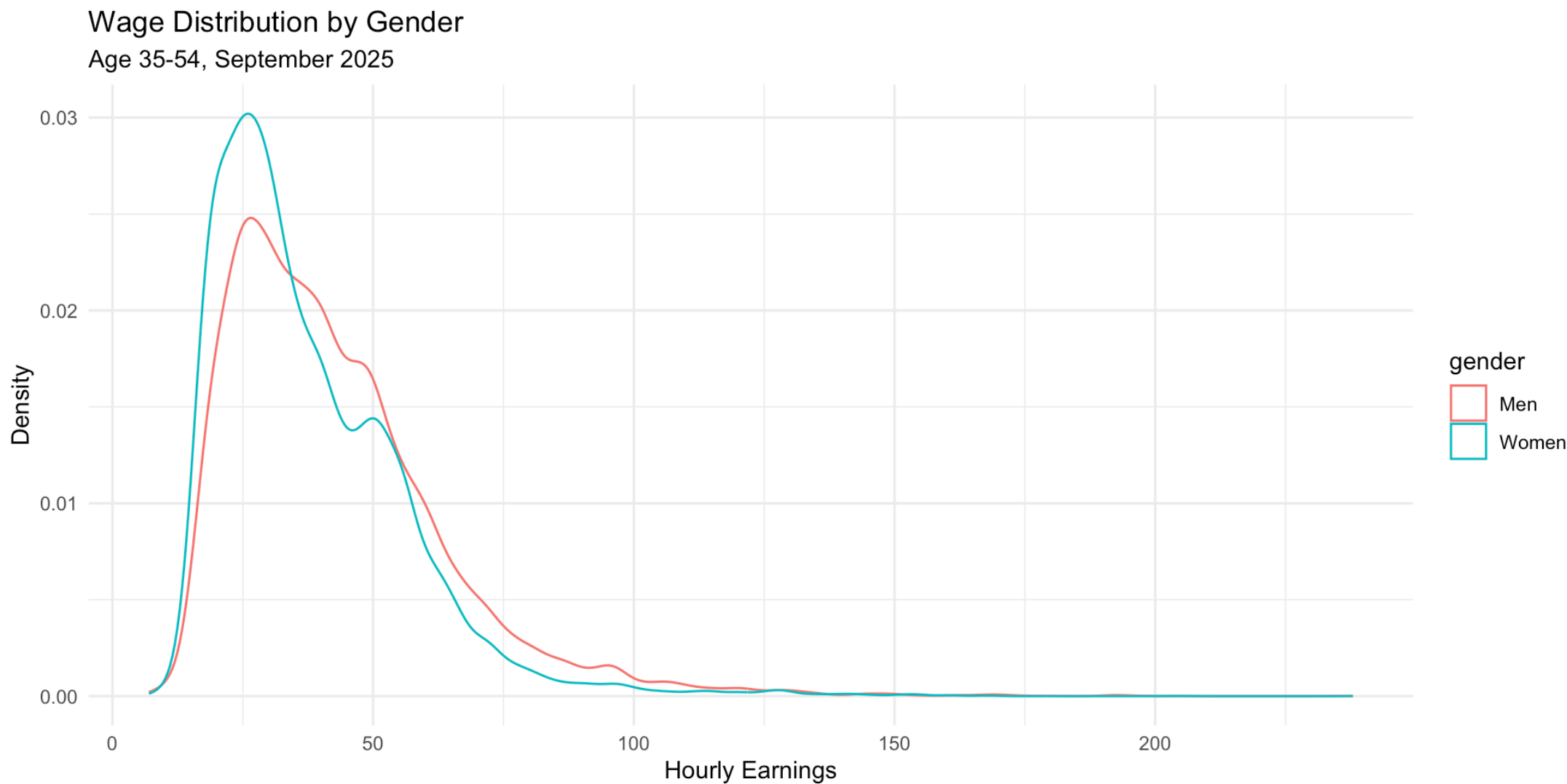
$$0.68 = \underbrace{0.16}_{\text{explained}} + \underbrace{0.52}_{\text{unexplained}}$$

- Interpretation:
- Explained: 0.16 (about 24% of the gap)
 - Due to men having higher average education and experience.
- Unexplained: 0.52 (about 76% of the gap)
 - Due to different returns to characteristics and intercept differences

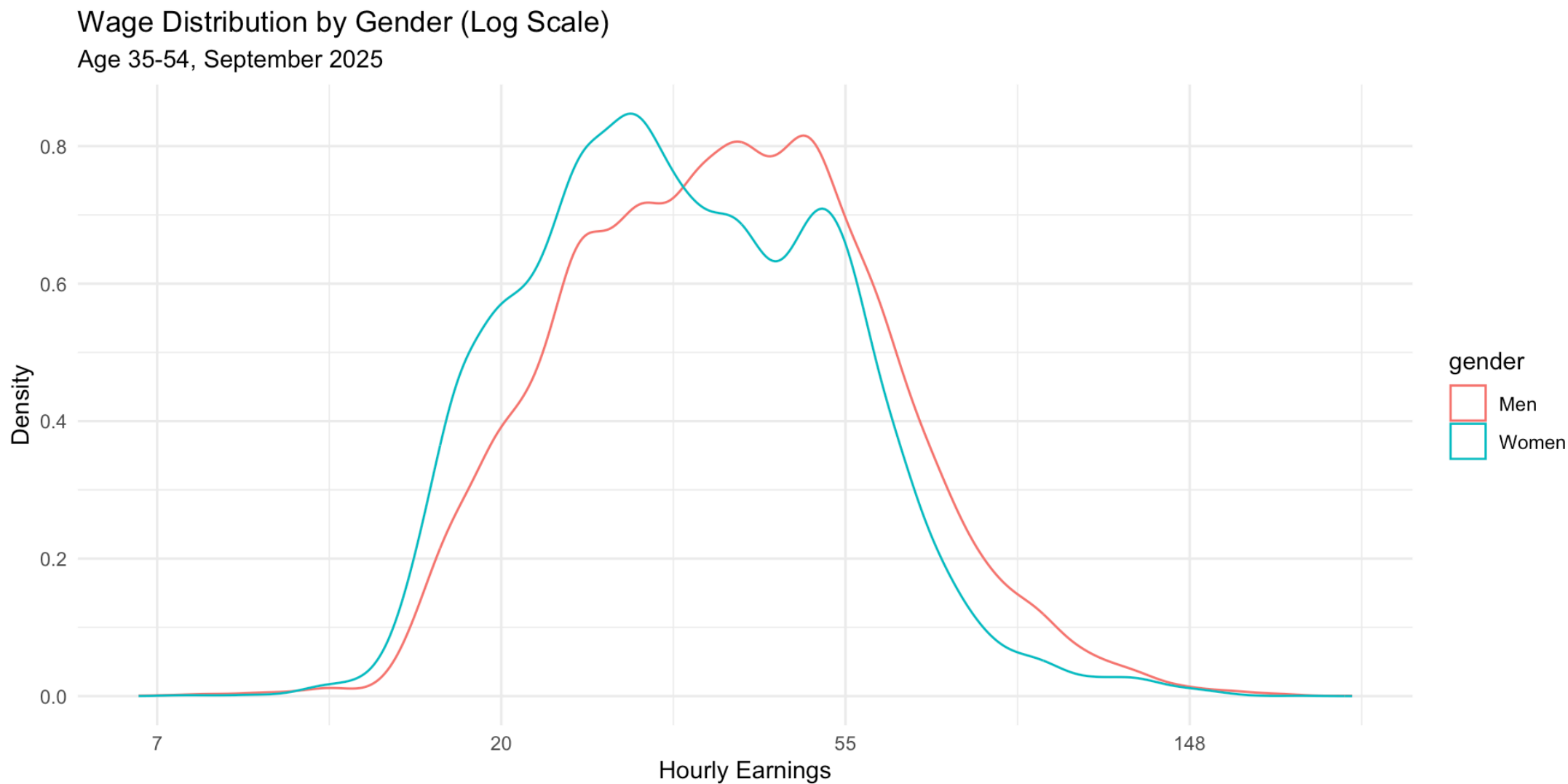
Blinder-Oaxaca Decomposition

- Here is an example using real data from Canada
 - Labour Force Survey from September 2025
 - Restricted to people aged 35-54
 - Have wages > 0
- First examine some summary statistics
- Then do a regular regression
- Finish with decomposition

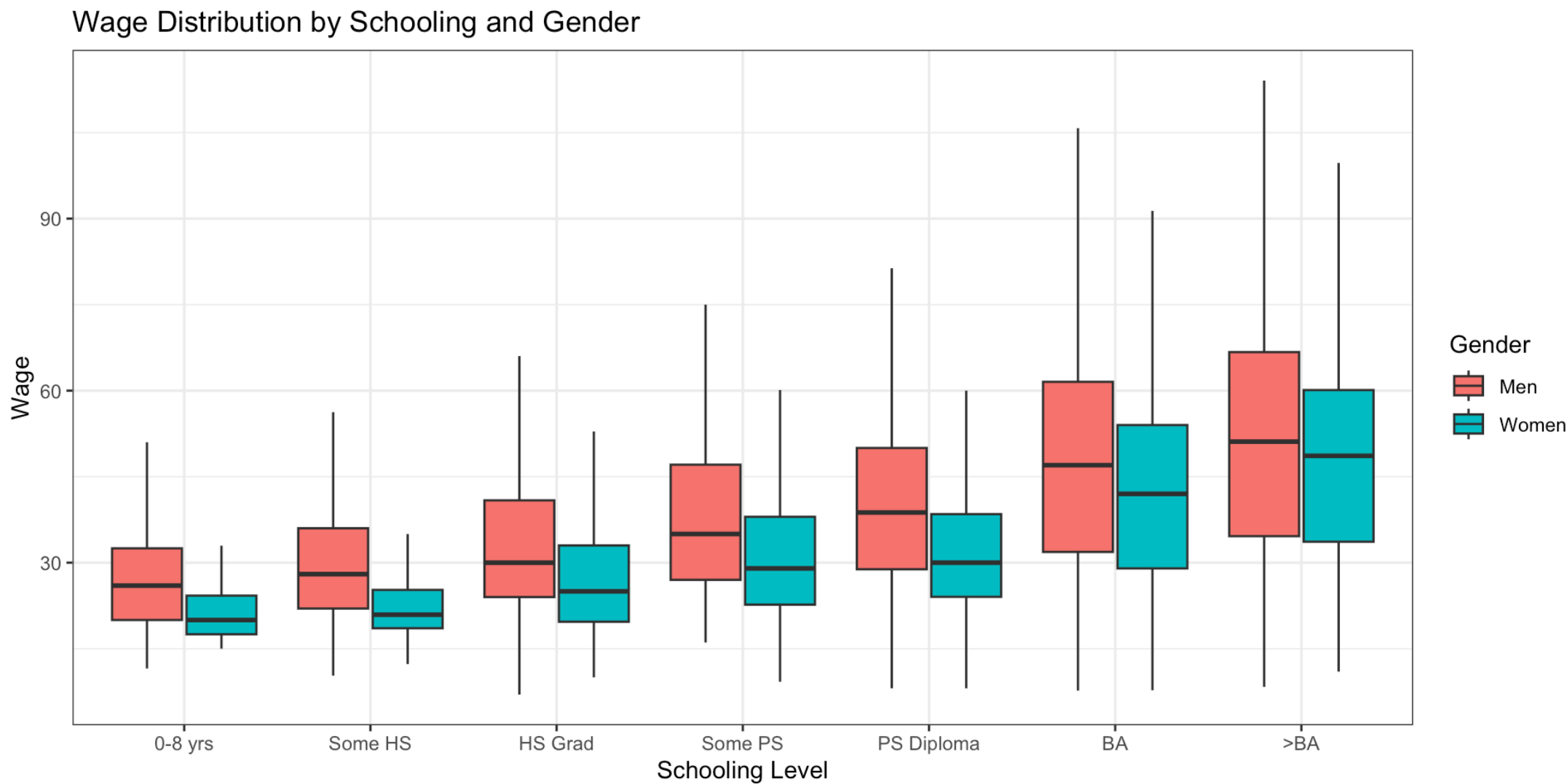
Blinder-Oaxaca Decomposition



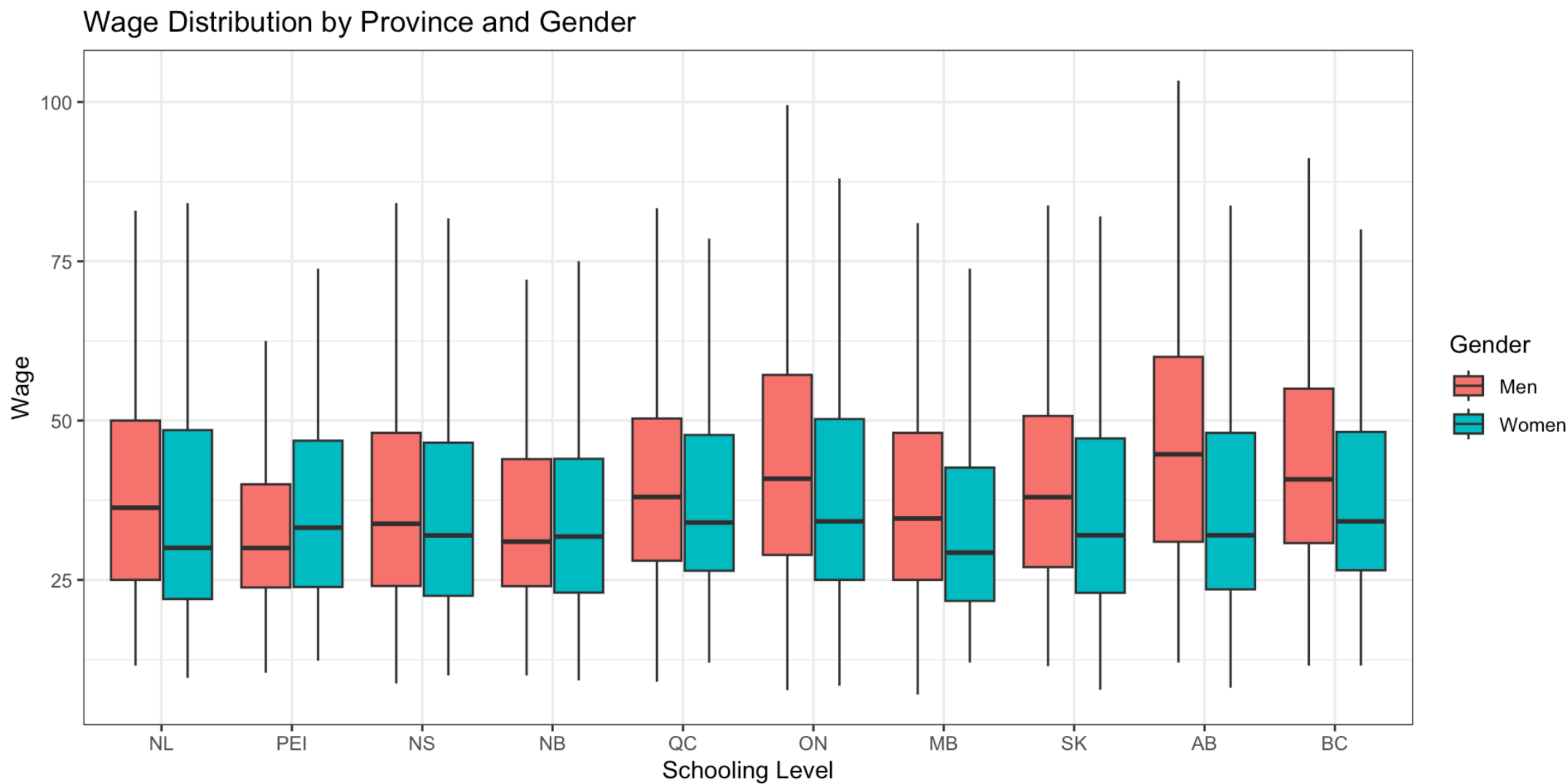
Blinder-Oaxaca Decomposition



Blinder-Oaxaca Decomposition



Blinder-Oaxaca Decomposition



Blinder-Oaxaca Decomposition

	Wage	Log Wage	Log Wage (with Controls)
Women	-5.032	-0.122	-0.174
	(0.247)	(0.006)	(0.005)
Num.Obs.	25233	25233	25233
R2	0.016	0.018	0.227
R2 Adj.	0.016	0.018	0.226
AIC	221896.5	212082.5	206091.9
BIC	221920.9	212106.9	206335.9
Log.Lik.	-110945.268	-15494.659	-12472.344
F	413.611	466.878	264.816
RMSE	19.65	0.45	0.40

Blinder-Oaxaca Decomposition

Component	Estimate
Overall Difference	0.122
Explained (Endowments)	-0.051
Unexplained (Coefficients)	0.173

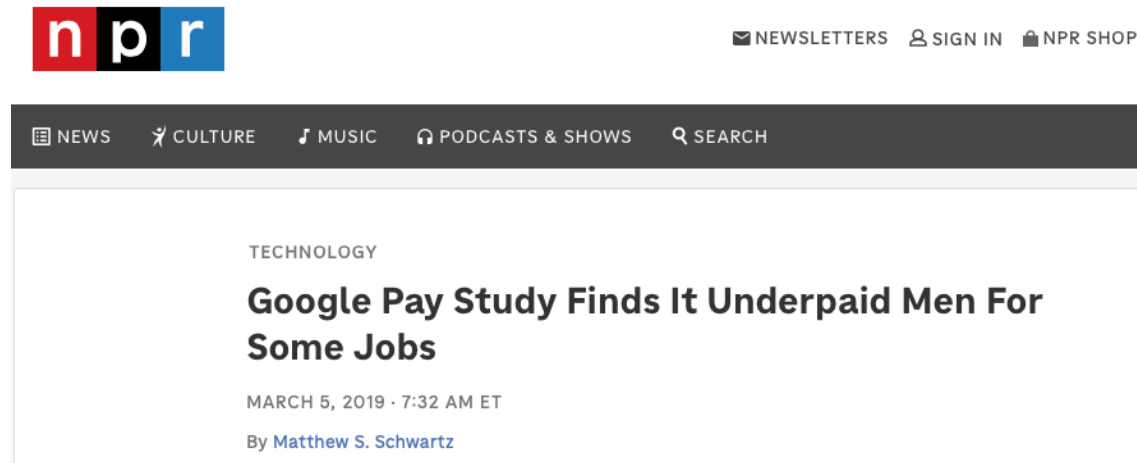
Collider Bias

- Sometimes researchers will control for variables in a regression to adjust the wage gap
 - What is the gap if we hold constant age, education, experience, occupation, industry, hours worked, etc.
- The idea is to compare people who are the same on all relevant characteristics
 - So that any remaining difference is (possibly) due to discrimination
- When doing this, you run into a tricky issue called **collider bias**
 - Controlling for certain variables can actually **create bias** in estimates

Collider Bias

- Easiest to understand with an example
- Suppose we examine the tech industry
 - Men and women have equal ability in the general population
 - Ability is positively related to wages
 - Women face discrimination in tech, so to get the job they must have very high ability
 - Men can enter even with moderate ability
- If you look at the wage gap in tech jobs, it might appear to favour women
 - In the same occupation, women will have **higher** ability because they had to overcome discrimination
 - So comparing men and women within occupation **creates** ability bias

Collider Bias



- This happened at Google several years back
- There was a complaint that women were being underpaid
- But they did an analysis and found that **men** were underpaid
- Was a perfect example of collider bias
 - They looked within jobs
 - Women of very high ability were likely selected into those jobs
 - Their wages were higher because of that

Collider Bias

- How to avoid collider bias?
- Do not add variables that are outcomes of discrimination to regression
 - Occupation is often an outcome of discrimination
 - Industry may also be an outcome
- Instead include only predetermined characteristics
 - Things that are not affected by discrimination
 - Includes mainly basic demographics, possibly education and experience

Additional Evidence on Gender Pay Gaps

Introduction

- There has been significant statistical work on this issue
- Why?
 - Its importance
 - Availability of data
- We will briefly review some of the evidence here
 - A slightly greater focus on Canadian studies

Evidence

- Across several studies, women in Canada earn about 60% of what men earn
- After controlling for characteristics, the gap shrinks to about 85%
- Pay gap is smaller in the public sector
 - Pay practices are more egalitarian in the government
- Gap has been falling over time
 - Many factors could contribute to this
- Internationally, gaps are smaller with stronger bargaining structures
- Gap is smaller the more competitive the industry

Evidence

- Occupational segregation does not explain a large part of the gap
 - Gap is explained more by *industry* rather than occupation
- Women face career penalties from family responsibilities
- Glass ceilings are an observed phenomenon
 - At the top end of the pay distribution, gaps are larger
 - Lots of international evidence of differences in career advancement
- Unions have a higher impact on women than men
 - Offset by the fact that women are less likely to be unionized

Evidence

- Mixed evidence on job protected leave
 - Leave is good because it protects jobs for women
 - But might make employers less likely to hire women in the first place
 - Evidence shows it lowers wages for women in the U.S.
 - In Canada it increases job continuity
- Some evidence that gap grows with more work experience
 - Upon initial entry, gap is small
 - Grows over time

Other Groups

Introduction

- The chapter is nominally about gender wage gaps
- But the same techniques can be used to estimate gaps for other groups
- This is usually done by race or ethnicity
 - U.S. studies have focused in particular on racial discrimination
- We will briefly cover this here

Evidence for Other Groups

- Indigenous peoples in Canada face significant wage gaps
 - About 85% of non-Indigenous wages off reserve
 - Shrinks after controlling for characteristics
- Some work on ethnic-white wage gaps in Canada
 - Complicated by the fact that many non-white Canadians are also immigrants
 - Immigration has its own independent effect on wages
 - Controlling for this, gaps are very high
 - Summarized on next slide

Evidence for Other Groups

TABLE 12.1

Ethnic Wage Differentials, Various Groups Born in Canada or United States

	Black	South Asian ^a	Southeast Asian ^b	Chinese	Aboriginal
Canada					
Total gap	0.293	0.536	−0.022	0.193	0.179
“Discrimination”	0.171	0.222	0.023	0.058	0.091
United States					
Total gap	0.328	0.176	−0.079	−0.205	0.304
“Discrimination”	0.123	0.055	−0.007	−0.047	0.136

NOTES: The total gap is the percentage by which the earnings of whites exceed those of the ethnic group, reflecting both the differences in their wage-determining characteristics and the differences in pay for the same characteristics. The latter is the differences in the regression coefficients from an Oaxaca-type decomposition as discussed in the text, and is labelled here as “Discrimination.”

^a E.g., Bangladesh, India, Pakistan

^b E.g., Philippines, Korea, Vietnam

SOURCE: Baker and Benjamin (1997, Tables 5 and 11, specification 3 including industry and occupation controls). Adapted from “Ethnicity, Foreign Birth and Earnings: A Canada/U.S. Comparison,” (Dwayne Benjamin with Michael Baker) in *Transition and Structural Change in the North American Labour Market*, Queen's University, Industrial Relations Centre and John Deutsch Institute, 1997. Used with permission.

Evidence for Other Groups

- Evidence is different in the U.S.
- Large literature on Black-white wage gaps
 - About 68% of white wages
 - Shrinks when controlling for past test scores (as measure of pre-market productivity)
 - Discrimination also coming from pre-market factors
 - Large component of discrimination
- Gaps favour Chinese and Southeast Asian workers
 - Not the same in Canada

Policies to Combat Gender Discrimination

Introduction

- Gender discrimination has been an issue for a long time
- Several policies are enacted to combat these
- Some examples:
 - Equal pay
 - Equal employment opportunity
 - Policies to facilitate female employment